Fair but hard: certified unique-solution puzzles generated by language models

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Abstract

Language models can write logic puzzles, but cannot check that a puzzle has exactly one answer, and a solver can only check a puzzle once it has been translated into constraints. We build a generate–encode–certify pipeline for four families (zebra puzzles in English, Sudoku, Killer Sudoku and Futoshiki) in which two independent model calls encode the text, a SAT solver proves the two encodings equivalent and the puzzle unique, and a ladder of five nested deduction engines grades how much reasoning each puzzle needs. Across 298 zebra puzzles, encoders made almost no error on plain template text, the smallest model erred on 18% of puzzles with harder wording, and 3–4% of puzzles paraphrased by a model were encoded wrongly, mostly because the paraphrase itself had changed a clue. Dual encoding caught every error that only one encoder made but none that both made; the uniqueness check caught all but one of those, a paraphrase that had changed a clue's meaning and still left a unique puzzle. Puzzles written by a model were all valid (100 of 100) but easy and redundant, and some Sudoku reused the layout of a well-known published puzzle. The ladder correlates strongly with solver effort and explanation size in every family, but three Claude models solved most fair puzzles regardless of level; the clearest failures were the smallest model's on 9×9 Sudoku (58 % exact), and they did not grow with level. Evolving fair zebra puzzles against the smallest model found no puzzle it failed; on Sudoku, evolution found puzzles it failed, but they were easy by the ladder and failed by a few cells. Putting several puzzles in one call changed measured accuracy more than puzzle level did, so all reported solving calls hold one puzzle. We release the pipeline, a benchmark of certified puzzles with levels and minimal explanations, and every intermediate result.

1 Introduction

A logic puzzle is a promise: there is exactly one answer, and it can be reached by reasoning rather than guessing. Generating puzzles that keep this promise is an old problem in procedural content generation [2, 24, 26]. For grid puzzles it is usually solved with a constraint solver: Sudoku, for example, is a small exact-cover problem that a SAT or constraint solver settles in milliseconds [14, 25, 28], and deciding whether a puzzle has a second solution is the standard way to certify uniqueness, even though the problem is hard in general [30]. Minimal puzzles and rated difficulty are well studied for Sudoku [16, 19], and step-wise explanations of why each cell value is forced can be extracted from minimal unsatisfiable subsets [1].

Large language models (LLMs) are now asked both to write such puzzles and to solve them. Grid-logic puzzles have become a standard probe of LLM reasoning [13, 27]. Two problems follow. First, a model that writes a puzzle in natural language cannot check that the puzzle has one answer, or that the answer it states is the right one. Second, any fixed benchmark built from such puzzles can leak into training data [7, 22]; dynamic benchmarks are one response [10].

Pairing an LLM with a solver addresses the first problem, and has been explored for logical reasoning [17, 18, 31] and for formalising mathematics [29]. But a certificate is only as good as the translation behind it. If the formal encoding misreads "directly to the left of" as "somewhere to the left of", the solver certifies the wrong puzzle. We therefore treat the translation itself as something to verify.

This paper builds and tests a generate–encode–certify loop for four puzzle families (zebra puzzles in English, Sudoku, Killer Sudoku and Futoshiki) and uses it for three purposes.

1. *Faithfulness.* Two independent LLM calls encode the same puzzle text into a small constraint language; a solver proves the two encodings equivalent or returns a disagreement. We measure how often single encodings are wrong, what kind of errors occur, and how many the dual check catches (E1). We also measure how often LLM-written puzzles are valid without the loop (E2).
2. *A difficulty ladder.* We grade every puzzle by the weakest of five nested deduction engines that solves it without search (unit propagation, single-constraint elimination, one- and two-level lookahead, search), alongside solver statistics and minimal-core explanation sizes, and relate the ladder to LLM accuracy for three model sizes (E3).
3. *Fair but hard.* We evolve puzzles with LLM-proposed edits under the constraint that they stay unique and solvable at L1–L3, with an LLM solver's mistakes as fitness (E4), and let an LLM invent

new genres that the pipeline compiles and certifies (E5).

All puzzles, certificates, ladder levels and explanations are released as a benchmark that can be regenerated freshly at any time.

Our main findings are mostly about limits. Encoders made almost no errors on plain template text. The errors we found came from harder wording (the smallest model counting "from the right" from the left), from an ambiguous template of our own, and from meaning drift introduced by a paraphrasing model. Dual encoding caught independent misreadings but not errors shared by both encoders; the uniqueness check caught all but one of those, a paraphrase that changed a clue and still left a unique puzzle. LLM-written puzzles were all valid but easy and highly redundant, and some Sudoku reused the layout of a well-known published puzzle. The ladder tracks solver effort closely, but current models solve most fair puzzles regardless of level, so it says little about LLM accuracy at these sizes; the clearest failures were the smallest model's on 9×9 Sudoku, spread evenly over levels and mostly a few wrong cells. Evolution against the smallest model found no fair zebra puzzle it failed, and on Sudoku only puzzles that are easy by the ladder. Finally, how a model is called matters: with several puzzles in one call, accuracy depended on the batch more than on the puzzle.

2 Methods

2.1 Puzzles as finite-domain constraint problems

Every puzzle in this study is a finite-domain constraint satisfaction problem with three parts: variables with small domains, *rules* of the genre that every instance shares, and *clues* that define one instance and can be removed. A rule or clue is either an all-different constraint or a table constraint, given by a predicate over the values of a small scope of variables. We compile the problem to CNF with the direct encoding: one Boolean $x_{v,a}$ per variable v and value a, exactly-one clauses per variable, pairwise at-most-one clauses for all-different (plus at-least-one clauses when the scope is a permutation, such as a Sudoku row), and one clause per forbidden tuple of a table constraint. Uniqueness is checked by solving, adding a clause that blocks the solution and solving again; the second call must be unsatisfiable [14, 30]. We use MiniSat 2.2 through PySAT [4, 9] and repeat every certificate with an independent model in OR-tools CP-SAT [20]. Both solvers agreed on every puzzle we certified.

Families. We use four families. *Zebra* puzzles have N houses and K categories with N values each (default N=5, K=4); clues are generated from 11 relation types (same person, different people, house position, not at a position, directly left of, next to, not next to, somewhere left of, exactly one house between, at an end, either–or) and rendered into English by templates with several phrasings each. Each zebra clue carries a canonical expression in the position DSL described below, which is the ground truth for its meaning. *Sudoku* is the classic 9×9 game. *Killer Sudoku* is played on 6×6 grids with 2×3 boxes; clues are cages (sum, digits distinct) and, rarely, givens. *Futoshiki* is played on 5×5 Latin squares with inequality clues between neighbours and optional givens.

Certified generation. For each family we draw a random solution, build a pool of clues that are true in it, reach uniqueness, and minimise. For zebra puzzles we start from the empty clue set and repeatedly add a pool clue that is violated by the current counter-example solution, which guarantees termination. For the grid families we start from the full pool. Minimisation removes clues one at a time in random order (givens first for Killer and Futoshiki) and keeps a removal whenever the solution stays unique, so every clue of a released puzzle is necessary. All randomness is seeded; every stage writes JSONL checkpoints and is resumable.

2.2 The difficulty ladder

We grade a puzzle by the weakest deduction engine that reaches the full solution without search:

L1 unit propagation on the CNF. In Sudoku terms this covers naked and hidden singles; for clues it covers using a clue once the other side is known.

L2 L1 plus single-constraint elimination: generalised arc consistency on each rule or clue taken on its own, together with the exactly-one constraints of its scope [15, 21]. This covers eliminations licensed by one clue, naked and hidden subsets inside one row or category, and cage combinations.

L3 L2 plus failed-literal probing: place a candidate value, run unit propagation, and eliminate the candidate if this yields a contradiction [8, 11].

L4 L2 plus two-level lookahead: place a candidate, then run unit propagation and failed-literal probing inside that hypothesis.

L5 none of the above suffices; the solver must search.

All rules are sound and monotone, so each engine has a unique fixpoint, and the level can be computed in one pass that enables stronger rules only when weaker ones are exhausted. We call a puzzle *fair* if it is unique and graded L1–L3. Unit tests check the ladder on hand-built instances that sit exactly at each of the five levels, on the classic Einstein puzzle, and on a published Sudoku with known answer.

Solver-side metrics. For each puzzle we record (i) the fraction of variables fixed after the L1 and L2 fixpoints (a "propagation backbone"; the full backbone of a unique puzzle is trivially every variable), (ii) MiniSat decisions, conflicts and propagations for finding the solution and proving it unique, (iii) for every

non-given fact, the size of a minimal set of clues that forces it, computed as a group minimal unsatisfiable subset over clue selectors by deletion [1, 12], and (iv) clue redundancy, the fraction of clues whose individual removal keeps the solution unique.

2.3 LLM calls

All LLM calls were made through the Anthropic Messages API with `claude-haiku-5-5`, `claude-sonnet-5-5` and `claude-opus-5-5` (the model identifiers returned by the API), adaptive thinking, the same effort setting (`high`) for every model, and an output limit of 128,000 tokens. A pipeline stage writes one task file per call; the file is sent as the only user message, with the instruction to reply with a single JSON object, and the reply is stored next to it. Calls share no context, and no tools are available to the model. For every call we log the exact input and output token counts, the stop reason and the message identifier (`results/ledger.jsonl`). Encoding and generation calls hold 10 items each, so items within such a call are not independent of each other, but different encoders of the same item never share a call. Every solving call in E3 and E4 holds a single puzzle: in a first run with 4–5 puzzles per call, models ran out of output budget or guessed on batches of hard puzzles, so that accuracy depended on the batch rather than the puzzle (Section 3); that run is kept in the repository but not used. 1 of the 672 calls behind the reported results reached the output limit (a Haiku Sudoku evaluation in E4, scored as wrong). An earlier version of this study, made when no API key was available, ran the same task files through isolated Claude Code subagents (264 calls, 22 of which used more tools than allowed); its outputs remain in the repository history, and only the abandoned E4 pilot reported below comes from it.

2.4 E1: dual-encoding verification

An encoder receives the puzzle text and a short description of a position DSL in which `pos('tea')` is the house number of a value and each clue is one Python Boolean expression over `pos`, integers, comparisons, `Abs`, `And`, `Or`, `Not` and `Implies`. The prompt gives three worked examples (same person, fixed position, next to) and leaves every other relation to the model. The same expression can be evaluated concretely (to build table constraints) or symbolically in z3 [3]. Two encodings A and B are equivalent when $\mathit{base} \wedge (A \oplus B)$ is unsatisfiable, where *base* states the domains and the per-category all-different rules. The same test against the generator's own expressions gives a ground-truth check for single encodings, at the level of the whole puzzle and of each clue.

We encoded 298 zebra puzzles in three text conditions: 100 with plain templates; 100 with "hard" templates that use indirect references ("the left-hand neighbour of the Dane keeps the cat"), counting from the right, negated orderings, conditionals, "neither ... nor" and "between"; and 98 whose plain-template clues Sonnet rewrote into natural prose under an instruction to keep the meaning exactly (two of 100 paraphrases were missing or had a different number of clues and were excluded). Each puzzle was encoded three times: twice by Sonnet in independent calls with different batch compositions (A and B), and once by Haiku (C). For every clue-level disagreement with the ground truth we assigned an error category automatically, by testing equivalence with a list of plausible misreadings, and then read examples of every category by hand.

2.5 E2: validity of LLM-written puzzles

Sonnet wrote 40 zebra, 20 Sudoku, 20 Killer and 20 Futoshiki puzzles together with their intended solutions, in batches of ten, under an instruction that each must have exactly one solution and be solvable without guessing. Zebra puzzles were dual-encoded as in E1 and kept only when the two encodings were equivalent. A puzzle is *valid* if it parses, has exactly one solution, that solution is the stated one, and it is fair. The solver loop then (i) returns the solver's feedback (a second solution, the clues the stated solution violates, or a minimal conflicting clue set) to a fresh Sonnet call for repair, and (ii) minimises the puzzle by clue removal. We also measured redundancy and searched for repeated Sudoku layouts.

2.6 E3: ladder, solver metrics and LLM accuracy

From the certified banks (1,500 minimal puzzles per family, plus 8,000 extra Sudoku drawn to look for rare high levels) we sampled up to eight puzzles per family and level. Minimal zebra, Killer and Futoshiki puzzles are almost never L1, so the L1 cells were filled by adding random true clues to minimal L2–L3 puzzles until they reached L1; these are marked as augmented. No minimal Sudoku above L3 occurred among 9,500 generated, and only four L5 Killer puzzles occurred among 1,500. The final set has 116 puzzles. Haiku, Sonnet and Opus each solved every puzzle, one puzzle per call. We score exact solution and cell accuracy (over non-given cells); an unanswered item scores zero. We relate level to solver metrics by Spearman correlation with bootstrap intervals [5], and to accuracy with a logistic regression of exact solution on level, family and model fitted on the family–model cells that are not at ceiling (a pooled fit is separated, since some cells have no errors), with puzzle-level cluster bootstrap for the level coefficient, and a linear mixed model of cell accuracy with a random intercept per puzzle [23].

2.7 E4: evolving fair puzzles against a solver

We ran a $(\mu+\lambda)$ evolution of zebra puzzles with $\mu = 8$. Each generation Haiku (the training solver) solves

every individual; fitness is one minus the mean cell accuracy over all of an individual's evaluations, so survivors are re-evaluated every generation and lucky scores average out; ties prefer fewer clues. Sonnet proposes two variants per parent as edit operations on the formal clue list (add, remove or replace a clue of a typed relation) after seeing the puzzle, its solution and the solver's mistakes. The pipeline applies valid edits, discards clues that are false in the solution, restores uniqueness with random true clues if needed, and rejects any child that is not fair (L>3). Held-out testing re-solves the initial and final populations with fresh Haiku calls and with Sonnet and Opus. After the zebra runs (a 5×4 pilot and a 6×6 run) found no solver errors, we ran the same loop on 9×9 Sudoku, where Haiku had failed in E3: the initial population is 8 random minimal L3 Sudoku (as for zebra, the initial population is drawn at L3), Sonnet proposes edits to the givens (add, remove or move a given, using the solution), children are repaired to uniqueness by adding givens from the solution and must stay at L1–L3.

2.8 E5: LLM-invented genres

Sonnet was asked to invent six new genres and to write each as Python code against a small API: a grid size, a function that adds global rules (all-different or table constraints over cells) and a function that, given a solution grid, lists candidate clues true of it with a checking predicate and a text. The pipeline compiles the code in a restricted namespace, draws random solutions of the rules with a SAT solver, builds a minimal unique clue set from the candidate clues, and grades the result. We generated twelve instances per genre. For a solving test we took, by a fixed rule, the genre whose certified instances reach the highest level (ties broken by the number of certified instances) and gave its four deepest instances, in one call, to each of the three models.

3 Results

3.1 Certified banks

Every puzzle in the four banks was certified unique by MiniSat and, independently, by CP-SAT, and every clue of every puzzle is necessary. Table 1 shows the ladder levels. Minimal puzzles rarely sit at L1: removing every redundant clue removes the easy routes, so the weakest engine that suffices is usually single-constraint elimination (L2) or failed-literal probing (L3). Minimal Sudoku is the exception, because with 9 digits per unit a few givens already create hidden singles; across 9,500 random minimal Sudoku none needed more than L3. The ladder separates the families: Killer reaches L4 in a fifth of the puzzles and L5 in four, zebra reaches L4 in about one in nine, and 5×5 Futoshiki never needs more than L3. Puzzles generated from the hard text templates have a similar level distribution to plain ones, because the templates change the wording, not the constraint.

3.2 E1: how often is a single encoding wrong?

Figure 1 shows one puzzle end to end: the text, the two Sonnet encodings (here identical clue by clue up to letter case, and proved equivalent), and the chain of forced facts with the minimal clue set behind each. Table 2 and Figure 2 give the error rates.

On plain template text the two Sonnet encoders made no error and Haiku one (a direction flip): 1 of 3×1,369 clue encodings was wrong. On hard templates Sonnet A produced 1 wrong puzzle, Sonnet B 0 and Haiku 18 of 100 (18% [11, 26]). On paraphrased text the encoders were wrong on 3–4 of 98 puzzles (Sonnet A 3% [0, 7], Haiku 4% [1, 8]). Apart from Haiku on hard templates these rates are low and their intervals are wide; the more useful result is the audit of what went wrong (Section 3.2).

Detection. The dual check flagged every puzzle on which exactly one encoder of a pair was wrong (40 of 40 such puzzle–pair cases). It missed every error that both encoders shared: on paraphrased text the two Sonnet encodings disagreed on no puzzle at all, although both were wrong on the same 3, and Sonnet A and Haiku shared one wrong encoding of a hard-template puzzle. Three of these four shared errors made the encoded puzzle lose its unique solution, so the uniqueness check rejected them. The fourth passed both checks: the paraphrase of puzzle `zebra-00148` said "two houses separate the German from the rabbit owner" for a clue that placed them two houses apart, all three encoders encoded the text as written (a distance of three), and the resulting puzzle still had exactly one solution, a different one. Across all 298 puzzles and three encoders this is the only wrong encoding that passed both checks (Figure 2b). Uniqueness works as a second detector because a minimal puzzle has little slack: changing the meaning of one clue usually removes the intended solution or admits a second one, but not always.

Audit. Every clue-level error was assigned a category automatically, by testing it against a list of plausible misreadings, and we read examples of every category (`results/e1/error_examples.jsonl`). There were four groups. (i) *An ambiguous template of our own* (15 clues, each misread by all three encoders): "At least one house separates X and Y" was generated for "X and Y are not neighbours" ($|p_X - p_Y| \neq 1$, which allows the same house), and all three encoders read it as $|p_X - p_Y| \geq 2$. Their reading is the natural one; the template is wrong. Because the clue is still true of the solution, no answer changed, and neither check can see the error; only the formal ground truth revealed it. (ii) *Counting from the right*: "the fifth house from the right" encoded as house 5 (Haiku 17 clues, Sonnet A

Table 1: Ladder levels of the certified banks (minimal puzzles). All puzzles were certified unique by two solvers.

Family	n	L1	L2	L3	L4	L5	clues (mean)
Zebra	1,500	0	360	967	173	0	13.6
Sudoku	1,500	738	214	548	0	0	24.3
Killer	1,500	0	582	600	314	4	8.4
Futoshiki	1,500	0	619	881	0	0	10.1
Sudoku (extra)	8,000	3859	1110	3031	0	0	24.3
Zebra (hard text)	300	0	57	198	45	0	13.2

Table 2: E1. Puzzles with a wrong encoding (in brackets: wrong clue encodings) for each encoder, and puzzles on which a pair of encoders disagrees (in brackets: puzzles with at least one wrong encoder in the pair that the dual check flagged / all such puzzles).

		wrong single encoding			dual check: disagree (caught/with err.)		
Text	n	Sonnet A	Sonnet B	Haiku C	A–B	A–C	B–C
Plain templates	100	0 (0)	0 (0)	1 (1)	0 (0/0)	1 (1/1)	1 (1/1)
Hard templates	100	1 (16)	0 (15)	18 (35)	1 (1/1)	17 (17/18)	18 (18/18)
LLM paraphrase	98	3 (14)	3 (14)	4 (16)	0 (0/3)	1 (1/4)	1 (1/4)

1). This one misreading accounts for most of Haiku's errors on hard templates. The dual check caught it wherever Sonnet read the clue correctly; on the one puzzle where Sonnet A made the same misreading, the uniqueness check caught it. (iii) *Direction flips and over-strong readings*, all by Haiku: "somewhere to the left of" encoded the other way round (3 clues), and "somewhere between X and Y" read as one order only (3 clues). (iv) *Paraphrase drift*: Sonnet, asked to keep the meaning exactly, rewrote "exactly one house between X and Y" as "two houses separate X and Y" in 3 puzzles. The encoders then encoded what the text said. This is an error of the text, not of the encoding, and dual encoding by construction cannot catch it. The clue-level counts in Table 2 also include 11 clues per encoder from three paraphrased puzzles whose paraphrase listed the clues in a different order. The clue-by-clue comparison then pairs each encoded clue with the wrong original one, but the encodings of these puzzles are equivalent to the ground truth, so they are not errors.

3.3 E2: LLM-written puzzles without and with the loop

Sonnet's puzzles were all valid (Table 3): 100 of 100 parsed, had a unique solution equal to the stated one, and were fair, so the round of solver feedback had nothing to repair. The zebra puzzles' two encodings never disagreed.

The puzzles were, however, easy and redundant (Figure 3). All 20 Sudoku had exactly 30 givens and were solved by unit propagation alone, and every Killer puzzle was L2. On average 0.73 of a Sudoku's givens, 0.76 of a Killer puzzle's clues and 0.82 of a Futoshiki puzzle's clues could each be removed without losing uniqueness; for zebra puzzles the figure was 0.27. Minimising the LLM puzzles with the solver raised their level (Sudoku: 4 of 20 reached L3; zebra: 17 of 40). The 20 Sudoku used only 12 distinct layouts of givens; 2 used the layout of the example puzzle on the English Wikipedia Sudoku page, though none reproduced that puzzle's digits. LLM-written puzzles are thus valid but not a good source of hard or novel instances, and reuse of familiar material is a real risk for benchmarks built this way.

3.4 E3: the ladder, solver effort and LLM accuracy

Ladder versus solver metrics. Within every family the ladder level tracks solver effort and explanation size (Figure 4, Table 4). The Spearman correlation of level with MiniSat conflicts was 0.95 (zebra), 0.74 (Sudoku), 0.91 (Killer) and 0.86 (Futoshiki); with the share of variables fixed by unit propagation it was between -0.84 and -0.91. The level is not a proxy for the number of clues: in Sudoku the correlation with the number of givens was 0.13, and in the other families harder puzzles had *fewer* clues. Mean explanation size rises with level, so harder puzzles are also those in which each fact depends on more clues at once.

Ladder versus LLM accuracy. Table 5 and Figure 5 give the exact-solution rates; every call held one puzzle. Opus solved every zebra, Sudoku and Futoshiki puzzle and 89 % of the Killer puzzles; its four misses were Killer puzzles at L1, L4 and L5 (two of the four L5 puzzles). Sonnet missed 10 puzzles spread over all four families and Haiku 17, most of them Sudoku (Haiku 58 % exact, Sonnet 88 %). Haiku's Sudoku failures did not depend on level: it solved 5/8 at L1, 4/8 at L2 and 5/8 at L3. Sonnet's Sudoku accuracy fell with level (8/8, 7/8, 6/8). Most misses were near misses: every wrong Sudoku answer of Sonnet had 98% of its cells right, and Haiku's Sudoku cell accuracy was 96.5 % overall; only two of Opus's Killer answers were far off (a quarter of the cells right or less). One item



Figure 1: One zebra puzzle through the pipeline. (a) The text given to the encoders. (b) The two Sonnet encodings in the position DSL; here they agree clue by clue up to letter case, and z3 proves $A \oplus B$ unsatisfiable. (c) Every fact of the solution in the order the ladder derives it, the strongest rule needed so far, and a minimal set of clues that forces it. The puzzle is L3: after one unit-propagation step every further fact needs failed-literal probing, and most explanations need 8–13 of the 14 clues.

Table 3: E2. Validity, level and redundancy of 100 puzzles written by Sonnet. "Valid" = parses, unique, matches the stated solution and fair (L1–L3). Levels are given as L1/L2/L3 counts before and after solver minimisation.

Family	n	unique	as stated	valid	after repair	levels (LLM)	levels (min.)	redundancy	clues
Zebra	40	40	40	40	40	1/32/7	0/23/17	0.27	15.9 → 14.0
Sudoku	20	20	20	20	20	20/0/0	9/7/4	0.73	30.0 → 22.6
Killer	20	20	20	20	20	0/20/0	0/19/1	0.76	16.4 → 9.1
Futoshiki	20	20	20	20	20	1/19/0	0/15/5	0.82	19.9 → 9.6

was unanswered (a Sonnet reply that was not valid JSON), and no call reached the output limit.

An earlier run of this experiment put 4–5 puzzles in each call. There, Haiku reached the 128,000-token output limit without answering on two of its five Sudoku calls (and one Killer call), and Sonnet answered others with grids that repeat a row pattern and break a given (Sonnet: 25% of Sudoku exact, against 88 % with one puzzle per call). Putting several hard items in one call thus changed measured accuracy far more than the puzzles' level did; we report only the single-puzzle run (the batched run is kept in `data/llm_tasks_archive/`).

Formally, on Sudoku, Killer and Futoshiki for Haiku and Sonnet ($n = 168$; Opus and zebra are at or near ceiling), the logistic slope of exact solution on level was -0.11 (95% cluster-bootstrap interval -0.53 to 0.31; $p = 0.61$). A linear mixed model of cell accuracy over all 348 attempts gave a level slope of -0.016 per level (SE 0.007, $p = 0.01$): cell accuracy falls slightly with level. Adding the level to a model with MiniSat conflicts, explanation size and number of clues did not improve the fit (AIC 138.4 without the level, 140.0 with it). The honest summary is that for current models these puzzle sizes are mostly too easy for the ladder to predict accuracy: it predicts solver effort well and LLM failure poorly, because there are few failures to predict and those that occur look like slips in carrying out a long solution rather than failures of deduction.

3.5 E4: evolving fair puzzles against a solver

Zebra. A pilot on 5×4 zebra puzzles (from the earlier, subagent-based version of this study; Section 2.3) and the main run on 6×6 zebra puzzles both ended without a single training-solver error. Over 3 evaluated generations the 6×6 run produced 22 fair individuals

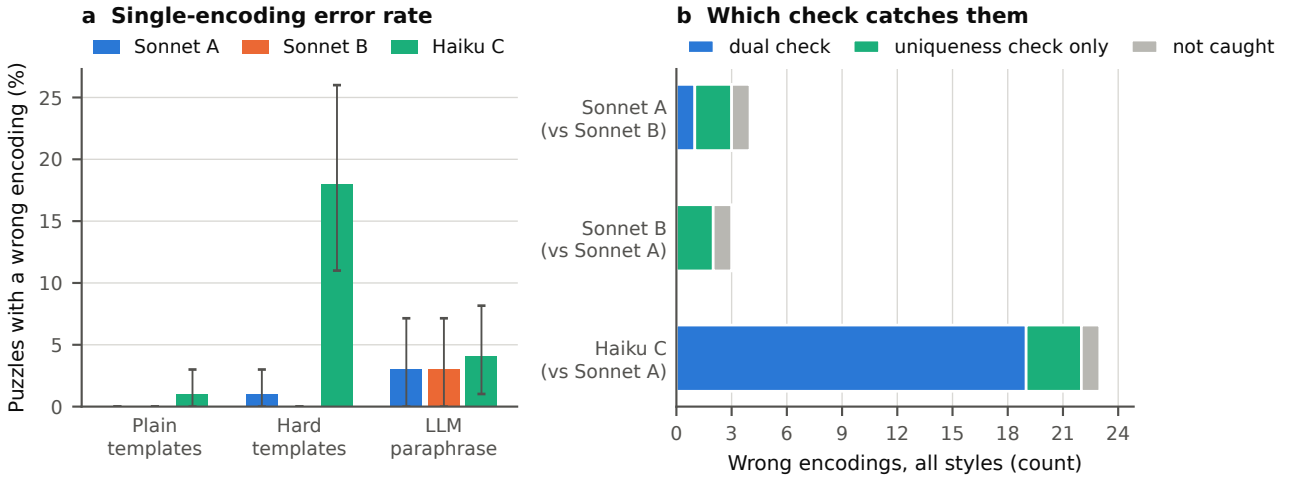


Figure 2: E1. (a) Share of puzzles with a wrong single encoding, with 95% bootstrap intervals. (b) For each encoder, how its wrong encodings were detected: by disagreement with its partner encoding, or only because the wrongly encoded puzzle no longer had a unique solution. One wrong encoding, made by all three encoders of a paraphrase that had changed a clue's meaning, escaped both checks (grey).

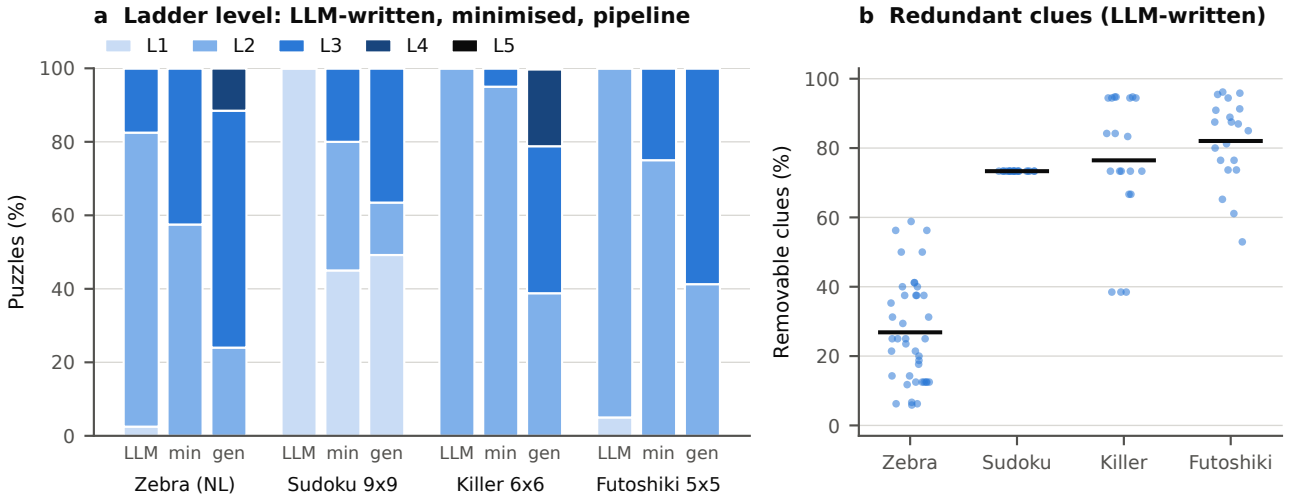


Figure 3: E2. (a) Ladder levels of the LLM-written puzzles, of the same puzzles after solver minimisation, and of the pipeline's own bank. (b) Fraction of clues that can each be removed without losing uniqueness; the pipeline's puzzles have none by construction.

(4 at L2, 18 at L3) from 14 accepted of 19 proposed edits, and Haiku solved every one of 38 evaluations exactly; the pilot (22 individuals, 30 evaluations) was the same. With no error, fitness is zero everywhere and selection has nothing to act on (Figure 6a). Fresh calls of all three models also solved every initial and final individual (Figure 6b). Fair zebra puzzles of this size are therefore not hard for Haiku, and we stopped the zebra runs.

Sudoku. On Sudoku the training solver did make errors, so selection had something to act on: over 5 evaluated generations, 26 of 48 individuals had at least one evaluation that was not exact. Every edit Sonnet proposed was applied and accepted (40 of 40), but the children were not deeper than their parents. Sonnet only ever removed givens (142 removals, no additions or moves); repairing a child to uniqueness then added about five givens back from the solution, and 39 of the 40 children were L1, whereas every initial individual was L3. Best fitness fell from 1.00 to 0.43 as lucky first evaluations were averaged out by re-evaluation, and mean fitness ended where it started (0.15 and 0.15; Figure 6a). Fresh calls solved 50 % (Haiku), 88 % (Sonnet) and 88 % (Opus) of the final population exactly, against 62 %, 88 % and 100 % of the initial one (Figure 6b; 8 puzzles each, so these differences are within noise). Seven of the eight final individuals are L1, and every wrong held-out answer had at least 86% of its cells right. Evolution against Haiku thus found Sudoku that it, and once Opus, gets slightly wrong, but they are easy by the ladder: what it exploited is unreliability in writing out a long and simple solution, not a lack of deductive depth.

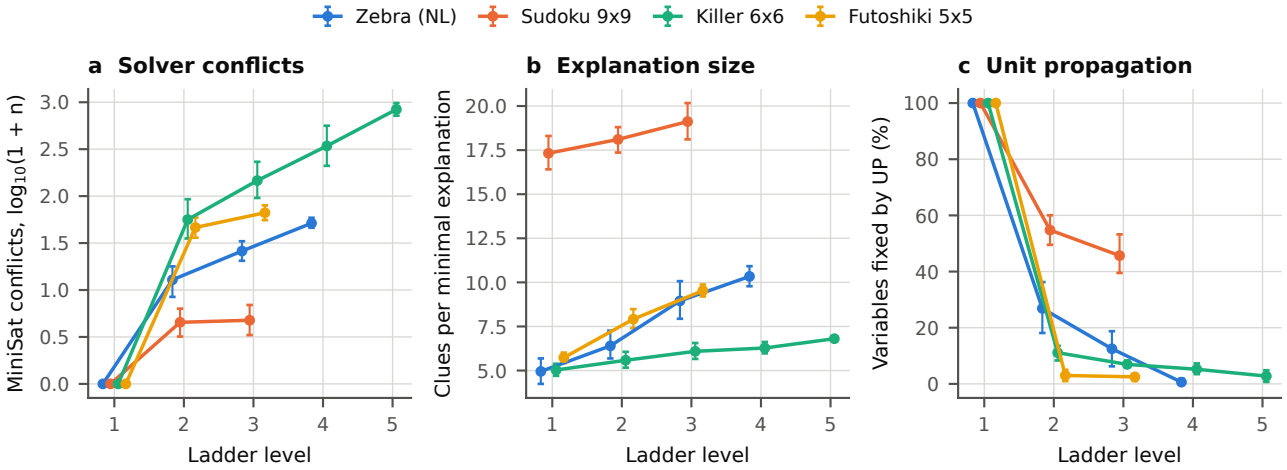


Figure 4: E3. Solver-side metrics by ladder level for the 116 selected puzzles (means with 95% bootstrap intervals). L1 cells of zebra, Killer and Futoshiki contain augmented puzzles.

Table 4: E3. Spearman correlation of ladder level with solver metrics, with 95% bootstrap intervals.

Metric	Zebra	Sudoku	Killer	Futoshiki
MiniSat conflicts	0.95 [0.87, 0.98]	0.74 [0.41, 0.93]	0.91 [0.80, 0.97]	0.86 [0.64, 0.97]
MiniSat decisions	0.92 [0.81, 0.97]	0.73 [0.39, 0.93]	0.92 [0.82, 0.97]	0.85 [0.62, 0.96]
mean MUS size	0.88 [0.78, 0.93]	0.45 [0.04, 0.77]	0.71 [0.49, 0.85]	0.91 [0.80, 0.95]
fixed by L1	-0.91 [-0.97, -0.80]	-0.84 [-0.96, -0.62]	-0.87 [-0.96, -0.72]	-0.77 [-0.94, -0.45]
probing steps	0.84 [0.70, 0.90]	0.84 [0.70, 0.89]	0.67 [0.39, 0.84]	0.84 [0.70, 0.89]
number of clues	-0.73 [-0.87, -0.48]	0.13 [-0.33, 0.57]	-0.77 [-0.91, -0.52]	-0.55 [-0.79, -0.07]

Table 5: E3. Exact solutions (%) with 95% bootstrap intervals, all levels pooled. Unanswered items count as wrong (one Sonnet zebra item whose reply was not valid JSON).

Family	Haiku	Sonnet	Opus
Zebra	97 [91, 100]	97 [91, 100]	100 [100, 100]
Sudoku	58 [38, 79]	88 [75, 100]	100 [100, 100]
Killer	83 [69, 94]	86 [75, 97]	89 [78, 97]
Futoshiki	100 [100, 100]	96 [88, 100]	100 [100, 100]

Table 6: E5. Genres invented by Sonnet: certified instances and their ladder levels.

Genre	certified	L1	L2	L3	L4	L5	clues
Latin Relations	12/12	0	0	10	2	0	9.9
Latin Windows	12/12	0	0	2	5	5	6.8
Knight Latin Peaks	12/12	0	1	7	4	0	3.4
Gapped Rows	12/12	0	4	8	0	0	7.1
Diagonal Echo	12/12	0	2	10	0	0	4.8
Even Windows	12/12	0	6	6	0	0	6.6

3.6 E5: genres invented by an LLM

All 6 genres that Sonnet wrote compiled in the sandbox, and the pipeline built and certified 12 minimal unique instances of each (Table 6). Most genres are Latin squares with a new kind of clue: relations between adjacent cells, statements about three cells in a line, a rule that cells a knight's move apart must differ, all-different diagonals, or 2×2 windows. Two change the base rules: one drops the column rule and instead requires vertically adjacent cells to differ by at least 2, and one uses a 4×4 grid with digits 1–5 and an even sum in every 2×2 window. Their certified instances are mostly L3. The deepest genre is *Latin Windows* (Figure 7b): a 5×5 Latin square whose clues describe 2×2 windows, giving a window's sum, which of its cells holds its largest or smallest digit, or whether a diagonal pair is equal. Ten of its 12 instances need two-level lookahead (L4) or search (L5), with about 7 clues. A window clue constrains four cells at once but fixes none of them, so the solver has to combine overlapping windows with the Latin rules, which is what the ladder detects. All three models nevertheless solved all four L5 instances of this genre that we gave them (12 of 12 attempts).

4 Discussion

What the solver loop buys. The loop turns a model's claim ("this puzzle has one answer") into a certificate, but the certificate covers the encoding, not the text. Our results suggest three practical rules. First, the translation from text to constraints is reliable when the text is regular: on plain template text the two Sonnet encoders made no error and Haiku one. Harder wording exposes systematic misreadings in the smaller model (Haiku counted "from the right" from the left in 17 clues), which a second, stronger encoder catches. Second, dual encoding catches independent

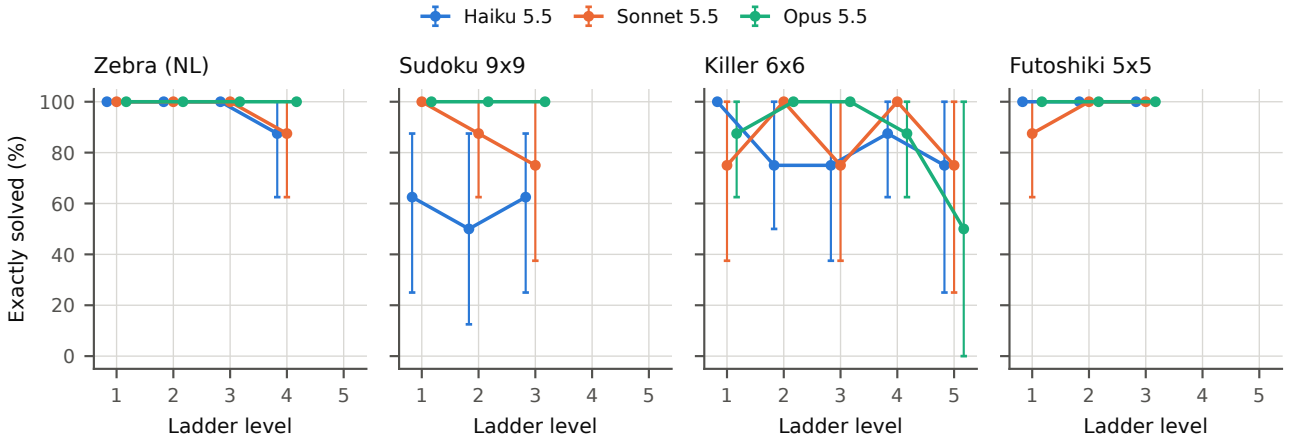


Figure 5: E3. Exact-solution rate by ladder level, family and model (95% bootstrap intervals; 8 puzzles per cell except Killer L5 with 4). Sudoku has no L4–L5 cells because none occurred among 9,500 generated puzzles; Futoshiki 5×5 has none above L3.

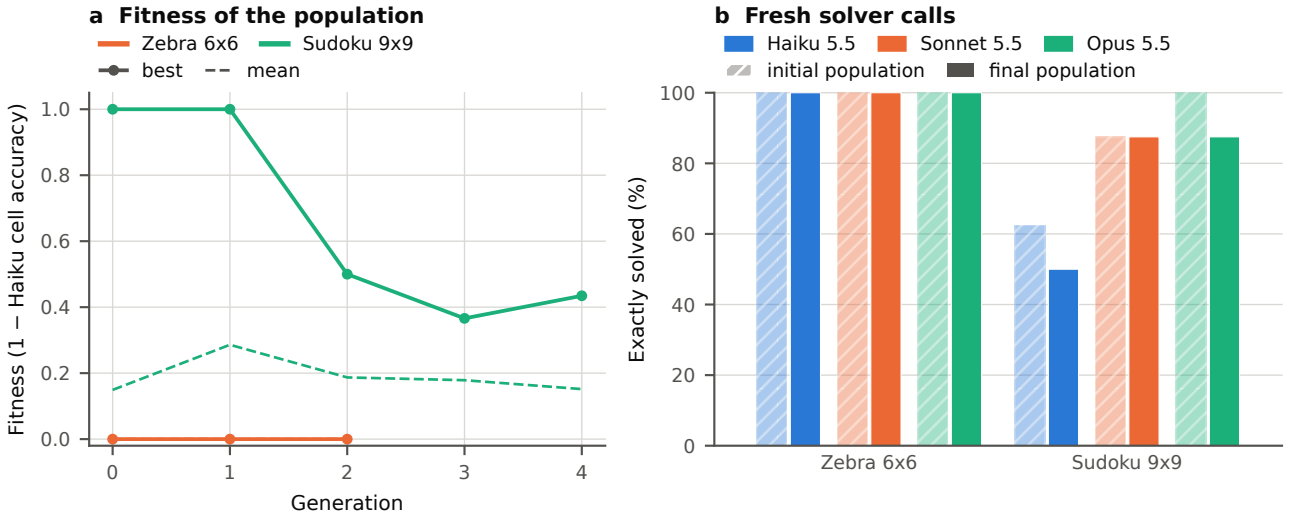


Figure 6: E4. (a) Best and mean fitness of the surviving population per generation for the 6×6 zebra and 9×9 Sudoku runs (fitness = 1 − mean Haiku cell accuracy over all evaluations of an individual; the zebra curves lie at zero). (b) Exact-solution rate of fresh solver calls on the initial (hatched) and final populations.

misreadings, which are what a second model run is good at, but not errors that two runs share, and two runs of the same model share a great deal: the two Sonnet encoders agreed on every paraphrased puzzle, including the three they both got wrong. Third, uniqueness is a strong and cheap second check for minimal puzzles, because a minimal puzzle has little slack: a misread clue usually removes the intended solution or lets in another. The combination caught every wrong encoding we observed but one. What it cannot catch is an error that leaves the puzzle unique: our own ambiguous template, which did not change any answer, and one paraphrase that changed a clue's meaning and still left a puzzle with a single, different solution. Such errors mean the text says something other than the certificate. The second kind is the more serious, because the released answer would be wrong; it arose in the paraphrasing step, which suggests certifying the text that solvers will read rather than an earlier version of it.

LLM-written puzzles. All of Sonnet's puzzles were valid, more than we expected. The problem is not validity but quality: they are easy, padded with redundant clues, and in Sudoku they repeat a few layouts, including that of a famous published puzzle. A solver that minimises and grades the puzzle fixes the first two; the third is a warning for anyone who builds evaluation sets by asking a model for puzzles.

The ladder. Five nested propagation engines give a level that is cheap to compute, interpretable in solving terms, and closely related to solver effort and to the size of minimal explanations in every family. It does not, at these sizes, predict well whether current LLMs succeed. They usually do, and when they fail the failures do not follow the level: the smallest model's Sudoku errors were spread evenly over L1–L3, and in E4 the Sudoku it got wrong were mostly L1. Most failures are a few wrong cells in an otherwise correct grid, which looks like unreliability in carrying out a

a Zebra, L3 (13 clues)

```
hobby: hiking, painting, golf, tennis, cooking
job: doctor, teacher, farmer, baker, pilot
drink: tea, coffee, juice, cocoa, water
color: blue, yellow, green, white, red

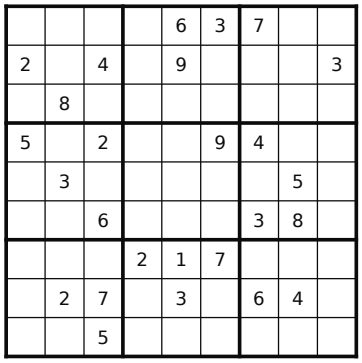
 1. The person in the green house lives in the fifth house.
 2. The person who enjoys cooking and the water drinker live
    in adjacent houses.
 3. The person who drinks water lives in the house just left
    of the one where the occupant of the white house lives.
 4. The tennis enthusiast lives directly to the left of the
    golf enthusiast.
 5. Whoever lives in the blue house and whoever lives in the
    yellow house live two houses apart.
 6. The tea drinker lives in the house just left of the one
    where whoever lives in the yellow house lives.
 7. There is exactly one house between the person who drinks
    coffee and the person who drinks water.
 8. The teacher lives in house 4.
 9. The water drinker lives immediately to the right of the
    baker.
10. The doctor lives in the last house.
11. The person who enjoys golf lives in house 5.
12. The pilot drinks juice.
13. The farmer and the person who enjoys hiking live two
    houses apart.
```

b LLM-invented genre "Latin Windows", L5 (6 clues)

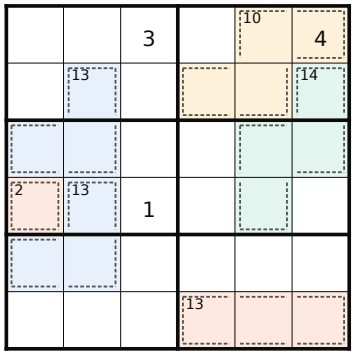
```
Fill the 5x5 grid with digits 1-5 so that no digit repeats
in any row or column. Clues describe 2x2 windows of the
grid: their sums, where their largest or smallest digit
sits, or whether a diagonal pair is equal.

- R1C5 = 5
- R5C5 = 4
- Window at R1C1 has sum 15
- R2C3 is the strict maximum of the window at R1C3
- Window at R3C2 has sum 14
- R4C4 is the strict maximum of the window at R4C4
```

c Sudoku, L3 (26 clues)



d Killer, L4 (9 clues)



e Futoshiki, L3 (11 clues)

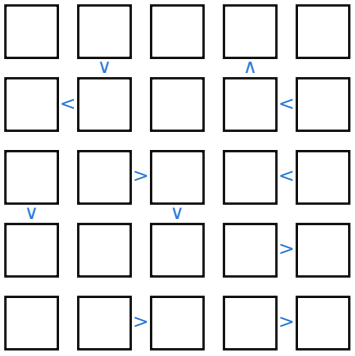


Figure 7: Gallery of certified puzzles from the benchmark: (a) zebra, (b) the LLM-invented genre Latin Windows, (c) Sudoku, (d) Killer (cages with sums), (e) Futoshiki. Levels are ladder levels; every puzzle is minimal and unique.

long solution rather than a missing deduction, and a measure of deductive depth does not capture that. How the model is called matters more than either: with several Sudoku in one call, the same models ran out of output budget or guessed, and accuracy fell far more than it does between levels. A ladder whose top levels are rare in random generation (no Sudoku above L3 in 9,500) is also a ladder whose top levels must be searched for, which is what E4 attempts.

Fair but hard. We did not find fair zebra puzzles that Haiku fails. This is a null result at our scale (two runs, at most three generations, 8 individuals) and it does not show that such puzzles do not exist; but the space we searched, all unique L1–L3 puzzles of size up to 6×6, was solved without error in 38 of 38 attempts. On Sudoku, evolution did find puzzles that Haiku fails, but they were L1 and failed by a few cells, so they are hard for Haiku in a different sense from the one the ladder measures; the repair step, which adds givens back, also pushes children towards easy puzzles. Larger instances, other genres, a repair step that preserves level, or fitness based on solving cost rather than errors may behave differently.

5 Limitations

- *How the models were called.* All calls used one setting per model (adaptive thinking, effort `high`, one sample per item; Section 2.3). Other effort levels or repeated sampling may change accuracy, and the batched first run shows how much the call setup can matter. The models' reasoning is not returned by the API, so we can describe failures but not inspect why they happened.
- *Scale.* The study is small: 298 puzzles in E1, 100 in E2, 116 in E3, three evolution runs. Rates near zero have wide intervals, and E3 has 8 puzzles per cell (4 for Killer L5); E4's held-out comparison has 8 puzzles per population.
- *Batching.* Encoding and generation calls hold 10 items, and E5's solving test four, so items in those calls are not independent. E3 and E4 solving calls hold one puzzle each.
- *Augmented L1 puzzles.* L1 cells of three families were filled by adding redundant clues to minimal puzzles, so they differ from the other cells in clue count and redundancy.
- *Ground truth in E1* comes from our generator; one

of its templates was ambiguous, which we report as a finding rather than an encoder error.

- *Cost* is computed from the exact token counts reported by the API at list prices, without batch discounts or caching.
- *Ladder.* The ladder is defined on one CNF encoding; a different encoding (for example with auxiliary variables) would change what unit propagation can do, and humans' notion of difficulty is not modelled.

Data and code availability

All code, configuration, prompts, task files, model outputs, intermediate results and figures are in the project repository, https://github.com/sarabehnamian/Puzzle-generation-with-guaranteed-uniqueness. The benchmark (`benchmark/`) contains the certified puzzles with their solutions, ladder levels, solver metrics and minimal-core explanations, with a datasheet [6]. Every number in this paper is written to `paper/numbers.tex` by `src/make_numbers.py` from saved results. The reported results use 672 Messages API calls with 8.4 million tokens, which cost $38 at list prices. Including the discarded batched solving run and a run affected by an answer-parsing bug that we fixed, the study made 971 calls with 16.2 million tokens for $63 (exact token counts; see `results/cost_summary.json` and `src/cost.py`).

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